Sanjivani Rural Education Society's

Sanjivani College of Engineering, Kopargaon-423603

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)

NAAC 'A' Grade Accredited

Department of Information Technology

UG Programme in IT NBA Accredited

NLP Mini PROJECT SYNOPSIS

(Academic Year 2026-2027)

Project Group Id	
Title of the Project	PolluAlert: An NLP-Based Pollution Emergency Detection and Alert System
Project Domain	Natural Language Processing (NLP) - Text Classification, Information Extraction & Environmental Incident Analysis
Problem Statement	Pollution complaints and incident descriptions are commonly reported through web forms, helplines, social media, news reports, and community observations. A large part of this information is available as unstructured natural language text. Existing pollution monitoring approaches mainly depend on physical sensors and manual complaint processing, which cannot directly analyse large volumes of textual reports or consistently identify important information such as pollution type, severity indicators, location, and possible emergency conditions. This project proposes PolluAlert, an NLP-based pollution incident analysis system that processes natural-language pollution reports and performs preliminary classification and information extraction. The system applies text preprocessing, TF-IDF feature extraction using unigrams and bigrams, and machine-learning classification using models such as Multinomial Naive Bayes, Logistic Regression, and Linear Support Vector Machine. It classifies reports into pollution categories such as air, water, soil/land, noise, industrial, waste/garbage, and chemical pollution, and provides preliminary severity and emergency-status classification. The system also extracts relevant locations, entities, and keywords and stores analysed reports in an SQLite database. A Streamlit-based interface is used to demonstrate the complete analysis workflow. The system is intended as a preliminary text-screening tool and does not replace environmental sensors, field inspection, or decisions by responsible authorities.
References	1. C. D. Manning, P. Raghavan, and H. Schütze, "Introduction to Information Retrieval," Cambridge University Press, 2008. 2. S. Bird, E. Klein, and E. Loper, "Natural Language Processing with Python," O'Reilly Media, 2009. 3. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011. 4. Central Pollution Control Board (CPCB), Government of India, "Pollution Monitoring and Environmental Information Resources." 5. Explosion AI / spaCy, "spaCy: Industrial-Strength Natural Language Processing in Python," official documentation.
Project Plan	*(separate sheet specifying all phases of project)*

Name of Students in the Project Group with their signatures	1. Abhishek Sopan Shelke 2. Pranav Prabhakar Shelke 3. Aditya Vikas Shende 4. Karnveer Nanasaheb Shinde
Name of the Guide and Signature	**Dr. S. A. Thanekar**
Date	

Project title Accepted/Rejected.

Dr. S. A. Thanekar
Project Coordinator

Dr. M A Jawale
HOD